

PhytoGraph — cycles 41-43

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PhytoGraph — cycles 41-43

Abstract

Cycles 41-43 preserved the terminal PhytoGraph closure state. The supplied audit report records no critical, moderate, or new minor findings. The worker correctly performed no build or run action because no new defect, reproducibility failure, public-facing issue, or qualifying scientific evidence was supplied.

The audit decision was `VALIDATED`. The master `prediction_ledger.tsv` and `speculation_ledger.tsv` each remained header-only with 1 line. `promise_ledger.jsonl` contained 236 events. `promise_check` and `org_check` both exited 0 with inherited nonblocking warnings only.

No track reopening predicate was met. All six tracks remain conservatively closed: Track 1 `sidecar_readiness_uncontrolled`, Track 2 `H2_remains_not_supported_or_data_limited`, Track 3 `confound_limited`, Track 4 `still_data_limited`, Track 5 `H5_remains_source_biased`, and Track 6 `environment_limited_untested`.

Introduction

PhytoGraph’s controlling scientific boundary is the master-ledger promotion rule. Track-local outputs, Atlas displays, closure summaries, and maintenance artifacts do not become campaign-level predictions unless they satisfy the validation and promotion predicates for the master ledgers.

Earlier reports established the substantive closure state. Cycles 33-35 consolidated the final free-tier six-track closure synthesis. Cycles 38-40 converted that state into a maintainer-facing handoff and repaired one local documentation link without changing scientific conclusions. Cycles 41-43 did not add a new repair, new scientific artifact, or new validation result. Their role was terminal maintenance: preserving the already validated non-promotion state.

Approach

This report consolidates completed work rather than re-auditing it. The primary source is the supplied cycles 41-43 audit report. Local workspace sources used for context were the final handoff manifest, the prior cycles 38-40 report, the current master ledgers, `promise_ledger.jsonl`, `REFERENCES.md`, and `MANIFEST.md`.

The supplied cycle-session IDs were:

Cycle	Researcher	Worker
41	793d9d8e-c320-4efa-96f6-ed5824aaa239	9d935685-0175-428f-b4bb-340dc19ccd6d
42	61171f32-595d-478a-b06e-9bba4a41f1f6	241545ec-920d-4584-b172-4721d562d031
43	71453875-320a-4ca3-aa91-f7910de2867f	bbfe9c09-98a3-43b2-853d-2fec8388eb14

Full native transcripts for these sessions could not be fetched in this environment because session-search/session-fetch tools were not available. That is a record gap. The report therefore relies on the supplied audit and local artifacts.

Findings

No-Op Maintenance Decision

The central finding for cycles 41-43 is that no action was warranted. The supplied audit states that the worker correctly performed no build/run action because the cycle input did not include a new defect, reproducibility failure, public-facing issue, or qualifying scientific evidence.

This is a terminal-maintenance result, not a scientific result. It does not add a prediction, falsification, source, benchmark, figure, schema change, Atlas change, or track-local output. It preserves the prior closure boundary.

Track Reopening Eligibility

The audit states that reopening eligibility was not met for any track. The six final track statuses therefore remain unchanged:

Track	Preserved status	Meaning
Track 1 Reticulation Atlas	sidecar_readiness_uncontrolled	Sidecar evidence remains insufficient for master promotion because accepted-key reconciliation and controls block upgrade.
Track 2 Ghost Hyperedges	H2_remains_not_supported_or_data_limited	No canonical held-out case passes the validation contract.
Track 3 Convergence Pressure	confound_limited	Trait evidence remains blocked by family-size and sampling-density confounds.

Track	Preserved status	Meaning
Track 4 Domestication Hypergraph	still_data_limited	Crop/CWR and occurrence scaffolds remain insufficient for recommendation-like climate substitution claims.
Track 5 Chemodiversity Predictor	H5_remains_source_biased	Temporal phytochemistry evidence remains source-biased and insufficient for promotion.
Track 6 Foundation Model Probe	environment_limited_untested	The static benchmark and scorer exist, but no qualifying local/open model response set was executed and scored.

Master Ledgers

The master ledgers remained non-promoted:

Ledger	Audit/local count	Status
prediction_ledger.tsv	1 line	Header-only
speculation_ledger.tsv	1 line	Header-only
promise_ledger.jsonl	236 lines	Append-only event ledger

The header-only prediction and speculation ledgers are the controlling evidence that no campaign-level prediction or speculation row was promoted during cycles 41-43.

Validators And Warnings

The supplied audit reports:

Check	Result
python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace	exit 0
python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace	exit 0

Both checks reported inherited warnings only. The audit characterizes these as historical nonblocking warnings, including legacy missing cycle/manager artifacts, old noncanonical paths, orphan final-report artifacts, and root-layout warnings.

Artifact Changes

No schema, evidence, Atlas, track, or ledger artifact was changed during the worker/auditor cycle work. For this reporter pass, MANIFEST.md was updated as the required current snapshot for report_cycles_41-43, preserving the existing ## Key Files section verbatim.

Discussion

Cycles 41-43 are a continuity window. Their significance is that the campaign did not drift after terminal closure. No track-local hypothesis was over-promoted, no master-ledger row was added, and no maintenance activity was used as a path to reopen scientific conclusions.

The audit guidance is explicit: future cycles should continue the terminal maintenance posture. Work should proceed only if a concrete maintenance defect, reproducibility failure, public-facing clarity/accessibility issue, or new qualifying scientific evidence is supplied with auditable validation and ablation controls.

This means the original campaign success criterion remains unmet. PhytoGraph remains a substantial infrastructure, validation, and falsification campaign, but not a campaign with one validated prediction per track.

Open Questions

The remaining questions are maintenance conditions, not active research questions:

- Will future cycles receive a concrete defect or evidence predicate that justifies action?
- Can inherited validator warnings remain documented as nonblocking unless a new run reports a new error?
- If new scientific evidence is supplied, will it satisfy the reopening predicates before any track status or master ledger changes?
- Can public-facing maintenance remain limited to clarity, accessibility, link repair, provenance hygiene, and reproducibility?

References

No external bibliographic references are cited for cycles 41-43. This report cites only local project artifacts and the supplied audit record.

Appendix: Implementation Details

Source Inventory

Source	Date / cycle	Contents	Timeline role
Supplied audit report	Cycles 41-43	Validation summary, no-op decision, ledger counts, validator outcomes, and guidance	Primary validation source
Cycle 41 sessions	Cycle 41	Session IDs supplied; full transcripts unavailable locally	Source gap
Cycle 42 sessions	Cycle 42	Session IDs supplied; full transcripts unavailable locally	Source gap
Cycle 43 sessions	Cycle 43	Session IDs supplied; full transcripts unavailable locally	Source gap

Source	Date / cycle	Contents	Timeline role
reports/cycles/report_cycles_38-40.md	Prior reporting window	Handoff manifest repair and communication-maintenance context	Immediate continuity source
reports/final_campaign_handoff_manifest.md	Final handoff	Canonical closure state, six track statuses, warning posture, reopening conditions	Closure boundary source
prediction_ledger.tsv	Current workspace	Header-only master prediction ledger	Non-promotion evidence
speculation_ledger.tsv	Current workspace	Header-only master speculation ledger	Non-promotion evidence
promise_ledger.jsonl	Current workspace	236 append-only events	Ledger-state source
MANIFEST.md	Current workspace	Updated current snapshot for future researcher/worker turns	Reporter-maintained artifact

Code Organization

No code was added or modified as part of the worker/auditor cycle work. Existing referenced maintenance code remains:

File	Lines	Purpose
scripts/build_taxonomy_results_site_assets.py	428	Builds the public taxonomy-results review site data, evidence tables, and figure assets from the final six-track status table.
tests/test_taxonomy_results_site_public_text.py	130	Verifies required site files, figure references, route labels, public-language boundary, six status codes, and header-only master ledgers.
reports/reopen/scripts/build_final_free_tier_closure_synthesis.py	197	Builds the final six-track free-tier status table, closure report, and summary figure consumed by handoff and site artifacts.
tests/test_final_free_tier_closure_synthesis.py	109	Verifies the final six-track status table, root/reopen propagation, figure presence, and header-only master ledgers.

File Counts

File	Count
prediction_ledger.tsv	1 line
speculation_ledger.tsv	1 line
promise_ledger.jsonl	236 lines
reports/final_campaign_handoff_manifest.md	76 lines
reports/taxonomy_results_site_closure_note.md	15 lines
reports/taxonomy_results_site_qa.md	40 lines
REFERENCES.md	157 lines
MANIFEST.md after reporter update	118 lines

Test Results

The reporter did not rerun the validator suite. The supplied audit reports these checks:

- `wc -l prediction_ledger.tsv speculation_ledger.tsv promise_ledger.jsonl`: 1, 1, and 236.
- `python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace`: exit 0 with inherited warnings.
- `python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace`: exit 0 with inherited warnings.

Cross-Reference Map

Origin	Consumer	Flow
Supplied cycles 41-43 audit report	This periodic report	Establishes the validated no-op terminal-maintenance decision.
reports/final_campaign_handoff_manifest.md	Future maintainer turns	Preserves final track statuses, canonical artifact links, warning posture, and reopening condition.
prediction_ledger.tsv and speculation_ledger.tsv	All closure records	Header-only state confirms no master prediction or speculation promotion.
promise_check and org_check outputs	Audit decision	Confirm validator success with inherited nonblocking warnings only.
MANIFEST.md	Future researcher/worker turns	Updated to summarize cycles 41-43 while preserving ## Key Files verbatim.

Record Gaps

Full native transcripts for the supplied researcher, worker, and auditor session IDs were not available because session-search/session-fetch tools were not available in this environment.

No new cycle-window scientific artifact was supplied or reported. The supplied audit explicitly says no schema/evidence/Atlas changes were made and no parallel fanout was warranted.

Coherence Review

One coherence pass was completed. The report defines the terminal-maintenance posture, distinguishes no-op validation from scientific progress, states the master-ledger boundary before applying it, and marks unavailable session transcripts as a record gap.